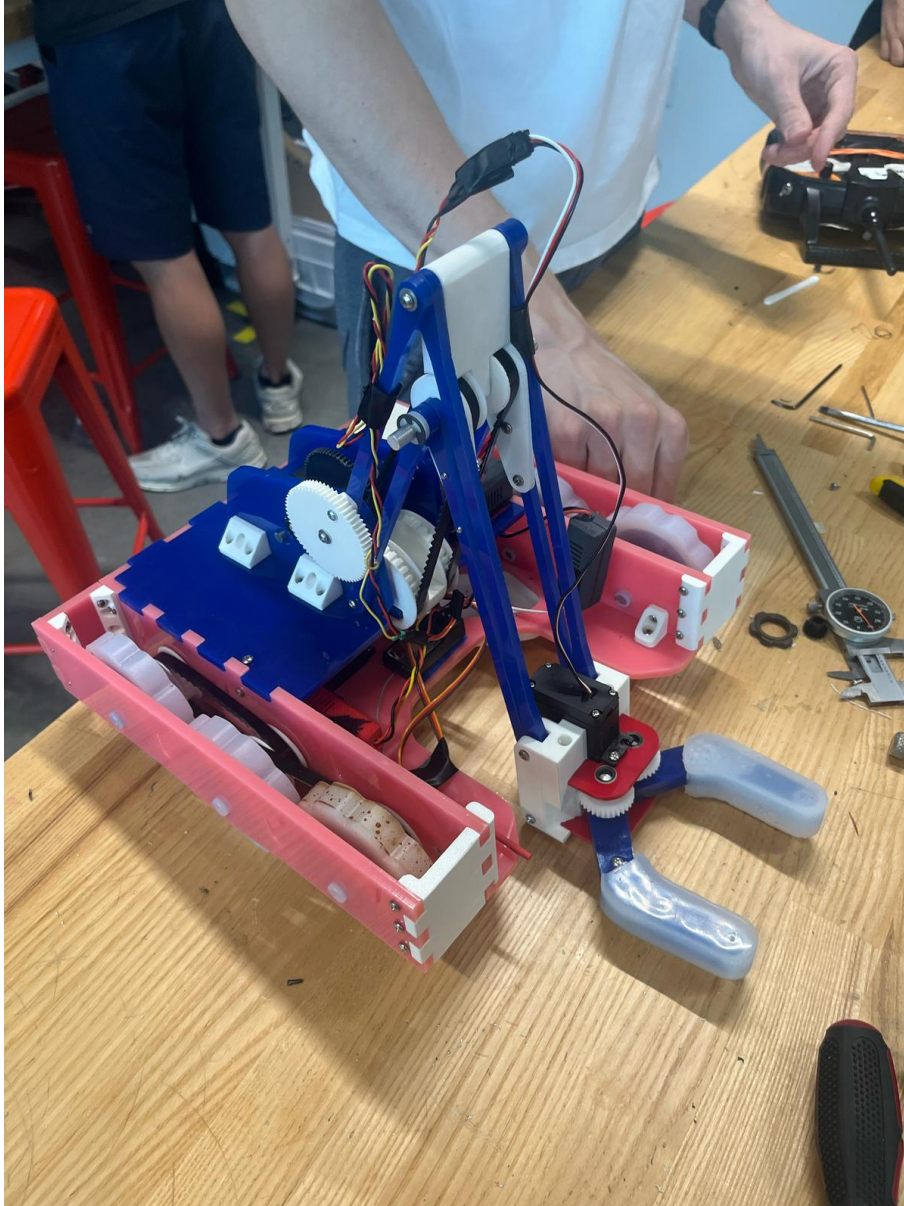


EngSci 51 Turf Wars Competition

Team 9: Seymour Galore



Submitted by:

Jack Kovac, Grant Kaufmann, Fotis Paganis, Abby Yoon, Sarah Li, Alex Ko

05.04.2024

Abstract

ES 51 assigned teams of 4-6 students to design, manufacture, and assemble robots to compete in a small field-based tic-tac-toe competition, requiring robots to pick up and move objects, as well as overcome small inclined planes. Team 9 (AKA Seymour Galore) aimed to design a robot that balanced maneuverability, high scoring potential, and adaptability to its opponents, giving it the greatest potential to win the competition. The final robot accomplished this in three facets: its innovative four-bar acrylic arm for extending pick-up reach, its versatile two-speed gear shifting six-wheel drive train for maximizing efficiency based on the terrain, and its flexible, servo driven silicone claw to easily pick up objects of various shapes and sizes. Each of these major assemblies was made using a combination of laser cut acrylic pieces, turned hex shafts using the lathe, PLA 3D printed parts, and silicone wheels and grippers cast from CNC machined molds, all manufactured by team members. Although Seymour Galore was eliminated early in the competition due to an arm failure, the robot's innovative 6 wheel drive train was frequently referred to as the most maneuverable on the field by competition moderators.

1 Concept Development

1.1 Field Parameters

This year's competition saw robots play 4 minute tic-tac-toe matches on a square, turf filled field against competing robots. Robots scored points by picking up scoring objects scattered throughout the field and depositing them in the tic-tac-toe grid. Robots could also gain additional points by getting tic-tac-toe.

1.1.1 Field Design

The square, turf field featured four corner ramps of blue and red colors. Two ramps were assigned to each team, with one being a 15° ramp and the other being a 30° ramp. As part of the competition, robots were required to successfully summit at least the 15° ramp.

1.1.2 Game Elements

The competition game elements were a square tic-tac-toe board in the center where robots would deposit game pieces and scoring objects. The three scoring objects were dog toys (worth 1 point), blocks (worth 1 point), and hockey pucks (worth 3 points).

1.2 Constraints

Several constraints were placed on the team and its robot throughout the design process. These constraints were taken into careful consideration as parts were being designed and manufactured, ultimately heavily influencing our final design.

1.2.1 Time Constraints

The competition build period ran from 02/06/25 - 04/29/25. The hard final deadline after which no more changes would be accepted was April 29th at 3:00 PM.

1.2.2 Material Constraints

Each robot was restricted to being made using provided stock parts, including 1/8 inch and 1/4 inch acrylic sheets, hex shafts, delrin blocks, extruded aluminum brackets, PLA spools of 3D printing filament, and various other supplied. Furthermore, tie-breaker winners were to be determined based on which robot had the lightest weight, meaning robots benefited from being made using less materials.

1.2.3 Actuator Constraints

Each robot could have a maximum of 6 actuators, maximum 3 of which could be motors (as opposed to servos). These motors came from small automatic screw drivers commonly found online while the servos were sourced from various hobbyist projects. Finally, the maximum allowed current draw was 9A.

1.2.4 Size Constraints

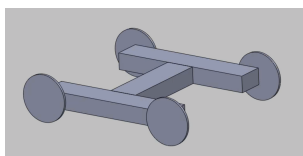
Every robot had to fit within the 11 inch by 11 inch by 11 inch "Box of Justice" at competition start. Robots were allowed to extend/move beyond the "Box of Justice" restriction once each match started, however.

1.3 Robot Brainstorming Designs

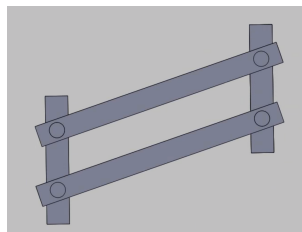
During design brainstorming, our team developed three initial models for the robot. Each of these models featured very different design principles outlined in the subsections below.

1.3.1 Design 1

This design featured an H or U shaped chassis with a vertical four-bar and a simple arm. Representing the most basic concept, this design had proven successful in past ES 51 competitions.



(a) H-Bar Chassis CAD



(b) Four-Bar Linkage CAD

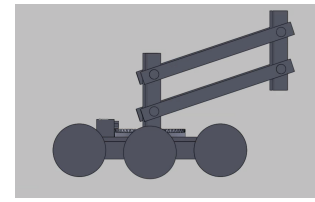
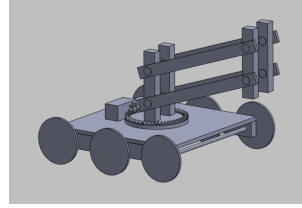
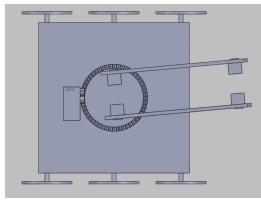


(c) Overall Prototype

Figure 1.1: Design 1 Prototype Elements

1.3.2 Design 2

The next model was a six-wheeled drive train with a pivoting vertical four bar. The end of the four-bar would have a claw attached to it for picking up game objects. Many elements of this design's chassis would end up in our final design due to the 6 wheel maneuverability, however the design as a whole had flaws due to its complex pivoting motion.

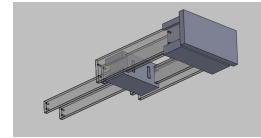
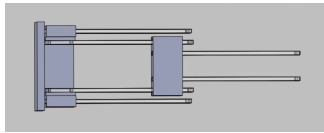


(a) Rotating Arm Top View CAD (b) Rotating Arm Overall CAD (c) Rotating Arm Side View CAD

Figure 1.2: Design 2 Prototype Elements

1.3.3 Design 3

Using the same chassis as Design 1, this design featured a telescoping arm. This arm provided the advantage of extending over the tic-tac-toe board to reach the middle square. Ultimately, the complexity of this arm design was its downfall.



(a) Linear Slides Top View CAD

(b) Overall CAD of Linear Slides Arm

Figure 1.3: Design 3 Prototype Elements

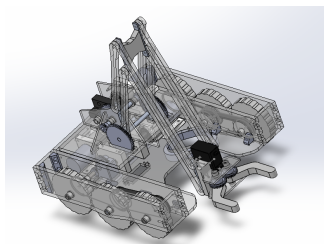
1.4 Design Selection

Our group utilized a Pugh Matrix (Table 1) to rank each design. The strengths of the final design (detailed in the next section) are reflected in Table 1 below.

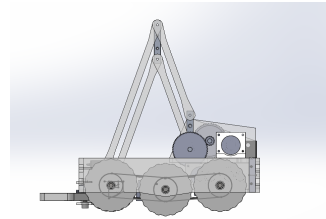
<i>Table 1: Pugh Matrix</i>						
Criteria	Baseline	Weight	Design 1	Design 2	Design 3	Final
Full Height	0	+1	-1	0	+1	+1
Design Simplicity	0	2	+1	-1	0	-1
Circuit Simplicity	0	1	+1	-1	0	0
Control Simplicity	0	3	0	-1	+1	+1
Intake Capability	0	3	0	+1	-1	+1
Deposit Design	0	2	-1	+1	0	+1
Total	-	-	-1	0	1	8

1.5 Final Design

Choosing the best aspects of each design, we landed on a six-wheeled, depressed middle wheel chassis with a horizontally extending four-bar. We determined that this design would allow for the most versatility moving around the field and while depositing game objects into the tic-tac-toe board. Next, members of our team created a full robot CAD very early on. These CAD designs influenced the future outcomes of how the robot was manufactured and assembled. Although it was revised throughout the engineering-design process, the fundamental tenets of the design remained constant throughout and are on display in the figures below.



(a) Isometric View CAD

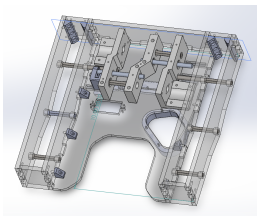


(b) Side View CAD

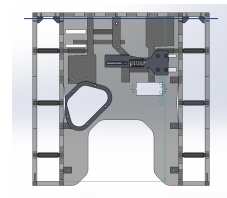
Figure 1.4: Final Design CAD

1.5.1 Chassis

Our final chassis was a two-speed, gear shifting, six wheeled design. The chassis was generally created in a U shape to leave space for a claw to pick up objects, with two sets of three wheels making up the sides and an acrylic plate holding the middle together. On each side, the middle wheel was slightly depressed. This design was intentional because it would allow the robot to pivot about that middle wheel with a smaller turn radius, thus maximizing maneuverability.



(a) Isometric View



(b) Bottom View with Gear Shifter

Figure 1.5: Final Design Chassis Assembly

Another unique element of this design, is the two-speed gear shifter featured on the robot's underside. This gear shifter allows the robot to move between two speeds during competition, one of which is the fast, driving speed, and the other of which is the slower, but higher torque

ramp climbing speed. Together, these two speeds ensure that our robot does not sacrifice torque or speed at a given moment, thus maximizing performance.

1.5.2 Arm

The final arm design represented the combination of various proposed designs from throughout the brainstorming process. Most notably, the design features a four-bar powered by a gear array attached to a screwdriver motor. It is important to note that this four-bar extends horizontally rather than vertically, taking inspiration from the telescoping arm of Design 3. The unique vertical nature of the four bar as shown below helps the arm stay within the "Box of Justice" requirements at the beginning of each match.

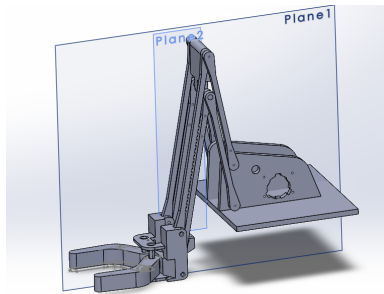


Figure 1.6: Final Design Arm Assembly

1.5.3 Claw

The final major aspect of the design was the claw. This claw featured two servo-powered prongs that were designed to grip game objects. Each of the prongs were designed to be surrounded by a rubber-like material (such as silicone) to improve grip on the desired object (whether that is a ball, a block, or a puck). To ensure that the claw fits within the "Box of Justice" at the start of each match, the claw was designed to leave room for the prongs to extend 180 degrees from each other so that they move back into the hole in the chassis, thus being well clear of the barrier set by the "Box of Justice."

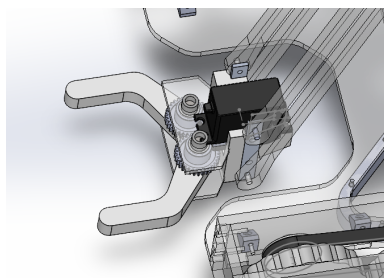


Figure 1.7: Final Design Claw

2 Analysis

2.1 Drive Train Calculations

Our drive train calculations focused on whether our chassis fulfilled our design specifications of what gear ratios we would need to maximize speed and torque for the dimensions of our robot. We took the overall mass of the robot to be 3kg. The wheel diameter to be 0.075m. We did center of gravity calculations both on a flat surface on our full assembly within Solidworks and found that the center of mass was 0.99 inches behind the center axel, and 0.89 inches above it. It was 0.1 inches off of the centerline for the full robot. Then, at max incline, 30 degrees, the center of gravity sits over the point in the axel plane. $0.99 + \sin(30) \cdot 0.89 = 1.43$ in behind the center axel. Since the axels are 3.2 inches apart, the center of gravity sits $3.2 - 1.43 = 1.77$ in in front of the rear axle, within the base of the robot ensuring stability on ramps.

We also completed torque calculations for our motors.

Gear Ratio and Drivetrain Calculations for 30° Incline

Motor Characteristics

V (V)	I_{stall} (A)	R (Ω)	T_{stall} (Nm)	No-load RPM	No-load I (A)	K_t (Nm/A)	K_v (RPM/V)
1.2	6.6	0.1818	0.302	90	2.0	0.0458	75
1.4	8.8	0.1591	0.344	105	2.2	0.0391	75
1.6	9.2	0.1739	0.410	115	2.2	0.0446	71.88

Estimated Parameters (from Curve Fitting)

Estimated R (Ω)	Estimated K_v (RPM/V)	Estimated K_t (Nm/A)
0.1716	73.96	0.0431

@6V:

Max Power Speed 221.88 RPM (3.698rps)

Max Power Torque 0.754Nm

Max Power 17.52W

When Restricted:

Max Current (I_{max}) 8A

Max Torque (T_{max}) 0.3451Nm

Drivetrain Calculations

Mass	3kg
Wheel Diameter	0.075m
Incline Force $F_x = mg \sin(30^\circ)$	14.7N
Required Axle Torque	0.5513Nm
Needed Gear Ratio (2 motors)	0.799
Linear Speed	1.091m/s

2.2 Arm Stress Calculations

Table 2.1: Mechanical Arm Parameters

Parameter	In/Lb	Parameter	Kg/M
Arm Length	17.0000	Arm Length	0.4318
Arm Mass	0.3800	Arm Mass	0.171
Payload Mass	1.0000	Payload Mass	0.153
Gripper Mass	0.3000	Gripper Mass	0.135
Beam Width	0.3750	Beam Width	0.009525
Beam Thickness	0.2500	Beam Thickness	0.00635
Beam Separation	2.0000	Beam Separation	0.0508
Pin Diameter	0.0625	Pin Diameter	0.0015875

In regards to calculations for the arm, we solved for the amount of stress that was on the arm to ensure it was below the yield stress of acrylic. The arm was 0.4318 inches in length, and had a mass of 0.171 kg. Additionally, the max mass for a game object was 0.153 so we completed our calculations off of that so that the arm would be able to lift all game objects. The mass of the class was 0.135 kg. The width of the beams for our arm was 0.009525m, and the thickness was 0.00635m. Finally, the beam separation was 0.0508m, and the pin diameter (diameter of the screw through the beam) was 0.0015875m. Additionally, there were 4 beams of acrylic per segment for the arm, so calculations were done assuming all four were in use at the same time.

$$\sigma = \frac{F}{A}$$

$$F = ma = mg = (0.171 + 0.153 + 0.135) \cdot 9.8 = 4.4982N$$

$$A = 0.009525 \cdot 0.00635 = 0.000060484 \text{ m}^2$$

$$\sigma = \frac{4.4982 \text{ N}}{4 \cdot 0.000060484 \text{ m}^2} = 18,593 \text{ Pa} = 18.593 \text{ kPa} = .018593 \text{ MPa}$$

Using these numbers, we calculated the normal stress in the beam to be 18.593 kPa.

The joint bearing stress can be calculated at the pins. The area of the bearing was taken to be the diameter of the pin times the width of the material: $.0015875 \cdot 0.00635 \cdot 4 = .00004032 \text{ m}^2$.

Then:

$$\sigma_b = \frac{F}{A}$$

$$\sigma_b = \frac{4.4982N}{.00004032 \text{ } m^2} = 111.556 \text{ } kPa = .111556 \text{ } MPa$$

Acrylic yield stress is between 44.9 - 86.0 MPa, which meant that our arm had a factor of safety of at least 400.

the amount of torque needed to raise the arm based on the dimensions of our robot and thus our gear ratio.

2.3 Arm Torque Calculations

We calculated the torque for the arm about the base where there was a joint connecting it to the robot. Taking the arm length to be 0.4318 m with an mass of 0.171. And the claw plus payload mass to be $0.153 + 0.135 = .288kg$, we can calculate the torque to be

$$.171 \cdot 9.8 \cdot \frac{0.4218}{2} + .288 \cdot 9.8 \cdot 0.4218 = 0.369189Nm$$

. We had previously calculated our max torque to be 0.345 Nm when restricted under 8 Amps of current, so we needed a slight gear ratio in order to guarantee our ar would lift. We needed a gear ratio of $\frac{.3692Nm}{0.3451Nm} = 1.069$. We selected gears with 64 teeth and 16 teeth in the first stage, and 64 teeth and 32 teeth in the second stage both to fit the dimensions of our design as well as provide us more than enough torque to lift the arm in a final gear ratio of 8:1.

3 Final Solution

3.1 Manufacturing

The process to get to our final product involved many hours of both additive and subtractive manufacturing.

Ultimately our chassis, the arm base, side plates, and wheel inserts were made of laser cut acrylic. In order to bolt everything together using M4-40 screws, every piece of acrylic was hand-tapped to make threads. Laser cutting acrylic came with the benefits of precision, but the drawbacks of acrylic being very brittle, so any part that had to be altered after laser cutting would be at risk of shattering.

We made heavy use of 3D printing. The motor and gear box mounts, battery holder, claw base, pulleys and gears, and other small parts were 3D printed. Due to failed prints and incorrect print settings, many parts had to be reprinted more than once. Some challenges we faced after printing were that tapped holes would easily strip out.

Another process we used was silicone molding. We used this to make the wheels, and the claw grips. It was a straightforward process, and we simply 3d printed our molds for the respective parts, filled the mold with silicone mixture, and then added inserts, and waited for the silicone to harden and removed the silicone.

The final manufacturing process that we made heavy use of was cutting our wheel and gear axles with the bandsaw. For many of our axles, we used steel hex shafts to mount the pulleys and wheels so they wouldn't turn around the axle. The ends were turned down using the lathe, from a hexagonal cross section to a circular one. They were turned until they spun inside of the nylon bearings with a running fit. We also used the grooving tool on the lathe to groove certain aluminum shafts and allow the insertion of a c clip to hold parts in place.

3.2 Final Assembly

The final assembly process came right down to the wire. As a result of being time constrained, we were unable to test all of the final components together. Some final issues that were addressed at the very end were the gearbox shifting mechanism, part of which had caused a gear to begin spinning about its axle. This required a last minute spring pin insertion to address the issue, and the gear box was permanently set to stay on one of the gear ratios. In addition, some of the screw holes did not align, or were inaccessible for a screwdriver, meaning some screws could not be inserted in the end. Some of the other screws, when tightened, exerted extra force on the motor and gearbox that caused them to not spin effectively. Ultimately our arm gears did not mesh effectively, which rendered the arm unusable.

3.3 Results/Discussion

By competition day, although we had fully assembled the robot, we did not have any time to test or even turn on any of the components of the robot. As a result, on competition day, we spent a significant amount of our time troubleshooting issues which would have come up immediately in testing.

Our first match, we failed to use any part of our robot. This was a result of "death by a thousand papercuts", that being a compounding of minor issues with tolerancing and mistakes made during assembly. Between the first and second matches, we spent some time sorting out each of the smaller issues with the robot. As a result, during the second match, we were able to have more success. Rather than attempting to score points, we played a defensive role, blocking the path of the other robot and preventing them from scoring. This was effective because although our robot had previously experienced issues, ours was quick on the field and was more reliable than the others. This is likely because we spent a considerable amount of time redesigning the gearbox that was provided to us, reducing the part count on the gearbox significantly and preventing it from experiencing the same issues that others did. This meant that we were successful in keeping the other team from scoring points, we just failed to win the weight competition.

Reflecting on our robot, and our build process overall, there are a few clear changes that should have been made. To start, we should have allotted more time for testing the final robot, even if it was only a few hours. Even in the 30 minutes between our competitions, we were able to debug a significant amount of issues with the robot. Had we done this in advance of the competition, we very well may have been able to win.

In terms of design, our 6-wheel chassis proved very effective, so much that it was one of the most reliable chassis on the playing field. While the shifter ultimately wasn't that effective during the competition, our robot was quite quick and maneuverable. At its core, the arm itself did not have any specific design issues, although there were a few issues with tolerancing. Had we taken additional time, this would have also been solved.

4 Appendix

4.1 Bill of Materials

Description	Quantity	Material	3D Print Volume (in³)	Manufacturing Technique
Base Between Side Plates	2	PLA	1.05	3D Print
Arm Motor Mount	1	PLA	2.60	3D Print
Gearbox Motor Mount	1	PLA	0.65	3D Print
Motor Shaft Mount	1	PLA	1.89	3D Print
Elbow Joint	1	PLA	1.31	3D Print
Bracket	2	PLA	0.10	3D Print
Claw Mount	1	PLA	3.81	3D Print
Long Bracket	4	PLA	0.18	3D Print
Short Bracket	5	PLA	0.10	3D Print
Shift Clutch	2	PLA	0.61	3D Print
Right Gearbox	1	PLA	3.50	3D Print
Center Gearbox	1	PLA	7.35	3D Print
Left Gearbox	1	PLA	3.57	3D Print
Battery Holder	1	PLA	1.48	3D Print
Shift Key	2	PLA	0.56	3D Print
Linkage Chassis	1	PLA	0.08	3D Print
Shift Pinion	1	PLA	0.07	3D Print
Linkage Arm	1	PLA	0.08	3D Print
Cylindrical Rings for Wheels	5	PLA	0.02	3D Print
Pinball Lever	2	PLA	0.32	3D Print
Arm Timing Belt Holder	1	PLA	1.144	3D Print
Claw Mold	2	PLA	2.17	3D Print
Total 3D Print Material (with 15% infill):			PLA: 38.374 in³	
Laser-cut				
Base	1	Acrylic ¼ in		Laser Cut
Side Plates	4	Acrylic ¼ in		Laser Cut
Back Plate	1	Acrylic ¼ in		Laser Cut
Arm Mount Plate	1	Acrylic ¼ in		Laser Cut
Motor Plate	2	Acrylic ¼ in		Laser Cut
Inserts for Wheel	6	Acrylic ¼ in		Laser Cut
Short Arm	2	Acrylic ¼ in		Laser Cut
Long Arm	2	Acrylic ¼ in		Laser Cut
Top Claw Mount	1	Acrylic ¼ in		Laser Cut
Bottom Claw Mount	1	Acrylic ¼ in		Laser Cut
Inserts for Claw	6	Acrylic ¼ in		Laser Cut

Machining			
Plate Axle	1	6061 Aluminum	Band Press
Hex Drive Shaft	1	Steel	Band Press & Lathe
Claw Axle	2	Steel	Band Press & Lathe
Short Gearbox Axle	1	Steel	Band Press & Lathe
Hex Gearbox Axle	1	Steel	Band Press & Lathe
Long Gearbox Axle	2	Steel	Band Press & Lathe
Base Round Shaft	1	6061 Aluminum	Band Press
Milling			
Wheel Mold	1	Polyurethane	Mill
Other			
Motor from Screwdriver	3	Gear 24T 32DP	8
Gear 32T 32DP	4	1277N26 Corrosion-Resistant Timing Belt Pulley	1
Gear 64T 32DP	1	1277N16 Corrosion Resistant Timing Belt Pulley	2
Gear 16T 32DP	8	Planetary Gearbox from Screwdriver	3
Spur Gear 64T	2	Servo Motor	1
1277N18 Timing Belt Pulley 18T	8	Servo Spring RC	1
1/4" Nylon Bearing	27	90295A140 Nylon Washer 1/4"	7
90126A505 Metal Washer Rings	4	90631A005 Locknut 4-40	2
Timing Belt .200" Pitch, 15" Outer Circle	2	Trimming Belt, 10" Outer Circle, .200" Pitch	2
Pan Head Phillips 4-40, 3/8"	15	Pan Head Phillips 4-40, 1/2"	32
Pan Head Phillips 4-40, 5/8"	10	97431A300, E-Clip 0.25"	2

4.2 Drawings

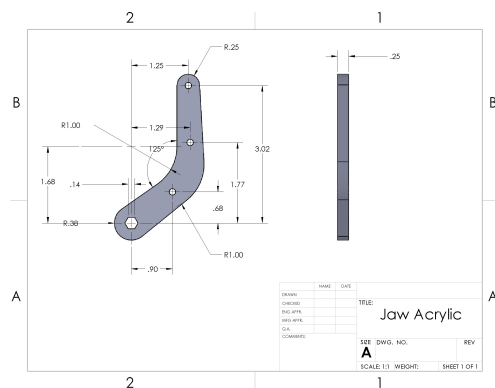


Figure 4.1: Acrylic Jaw

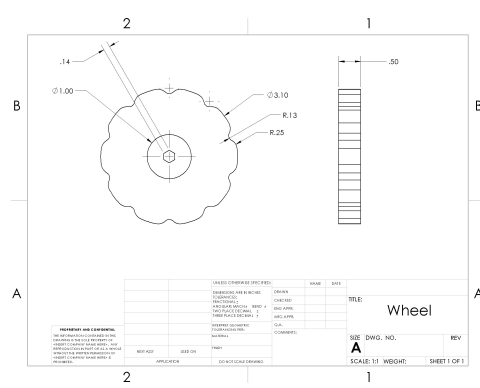


Figure 4.2: Wheel

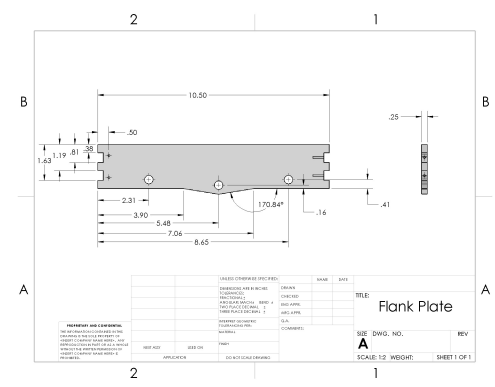


Figure 4.3: Flank Piece

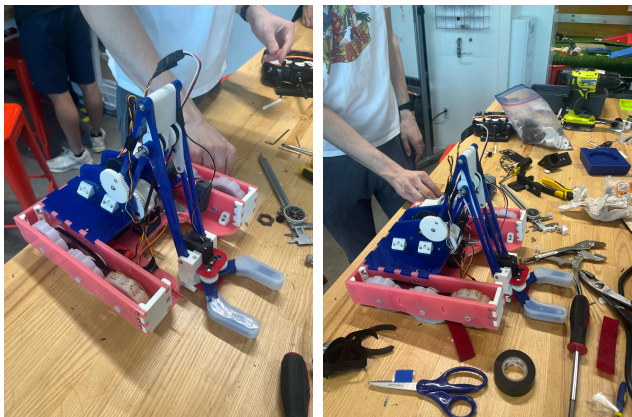


Figure 4.4: Full Robot Figure 4.5: Full Robot

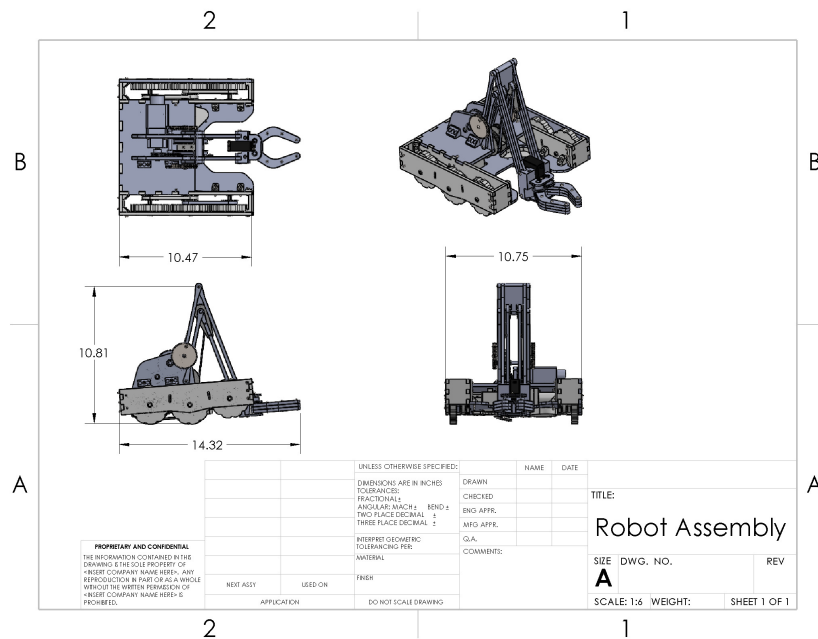


Figure 4.6: Robot Drawing

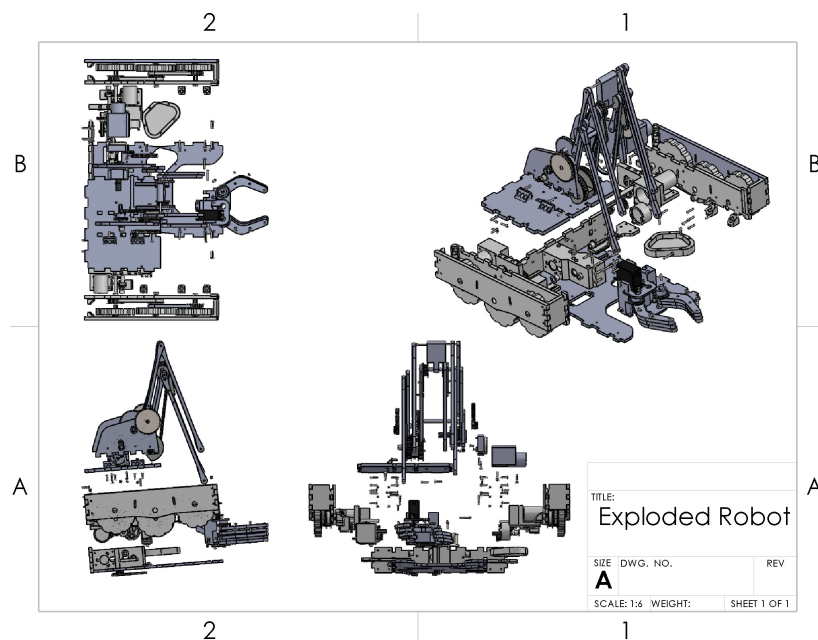


Figure 4.7: Exploded Robot Drawing

4.3 Credits

Within this report, Alex wrote the abstract and discussed design selection, Abby constructed the Bill of Materials, Sarah and Grant wrote the calculations, Grant, Fotis, and Jack worked on final solution and exploded views. Other parts were joint efforts to finish.